Element E

Application of STEM principles and practices

The proposed solution is well-substantiated with STEM principles and practices applicable to all or nearly all design requirements and functional claims; there is substantial evidence that the application of those principles and practices by the student or a suitable alternate has been reviewed by two or more experts (qualified consultants and/or project mentors) and that those reviews provide confirmation (verification) or detail necessary to inform a corrective response.

Prior to deciding on the twin drum, we had concluded that the omni-sphere would be the best option for the puzzle project, as it would provide the best defense against the strength of the bears as well as providing an enriching and fulfilling pass time for the bears. However as we researched possible ways of constructing the omni-wheel, we decided to change to the twin drum for cost effectiveness.

The omni-wheel was a design consisting of 2 spheres, an inner and outer. The inner shell would be surrounded by 2 ball bearing tracks letting the sphere have full mobility inside. Both the inner and outer shell would have a small hole where food or treats would be placed. The inner sphere would be weighed down on the side with the whole with sand or other harmless material. The trick for the bears would be to balance the two holes where gravity forces the treats out of both holes.

Our first reasoning for choosing the Omni sphere was because of strength. A spheres strength is primarily attributed to their geometric properties and material composition. Spheres distribute external forces evenly across their surface, making them inherently strong and resistant to deformation. This is because the stress is evenly distributed in all directions from the center of the sphere, resulting in a structure that can withstand significant pressure without collapsing.

Additionally, the surface tension of a sphere contributes to its strength by evenly distributing forces along its curved surface.

Furthermore we found a material that would work for our intended use, that being HDPE. High-density polyethylene (HDPE) is an excellent choice for providing the sphere with strength and safety for use with the bears due to its unique material properties. HDPE is a thermoplastic polymer known for its high strength-to-density ratio, making it lightweight yet remarkably strong. It is resistant to impact, chemicals, and moisture, making it suitable for outdoor and animal-related applications. HDPE is also non-toxic, which is crucial for ensuring the safety of animals interacting with the sphere.

High-density polyethylene (HDPE) exhibits remarkable strength due to its molecular structure and material properties. Studies have extensively investigated the mechanical properties of HDPE, providing empirical evidence of its strength and durability. One study published in the Journal of Applied Polymer Science examined the tensile properties of HDPE using standardized testing methods. The results revealed that HDPE exhibits high tensile strength, typically ranging from 20 to 37 MPa (megapascals), depending on the specific grade and manufacturing process. Additionally, HDPE demonstrates excellent elongation at break, often exceeding 600%, indicating its ability to deform significantly before failure.

The strength of HDPE stems from its molecular structure, which consists of long chains of hydrocarbon molecules with minimal branching. This linear molecular arrangement allows for strong intermolecular forces, particularly van der Waals forces, which contribute to the cohesion and resilience of the material. Furthermore, the absence of double bonds in the polymer chain enhances HDPE’s resistance to chemical degradation, ensuring long-term durability.

Furthermore, HDPE is easy to mold into complex shapes, including spheres, allowing for the creation of customized designs tailored to the specific needs of our project. Its smooth surface reduces the risk of injury or abrasion, providing a safe environment for the bears to interact with the sphere. Additionally, HDPE is durable and resistant to UV degradation, ensuring that the sphere maintains its strength and integrity over time, even when exposed to harsh environmental conditions.

Ball bearings are crucial components in machinery, facilitating smooth motion by reducing friction between moving parts. At their core, they operate on the principle of rolling friction, where the contact between the rolling elements and the races (tracks) results in less friction compared to sliding friction. This is governed by the laws of physics, particularly Newton's laws of motion. The balls or rollers within the bearing distribute the load evenly, allowing for efficient transfer of force. Additionally, ball bearings leverage Hertzian contact theory, named after Heinrich Hertz, which describes the behavior of elastic bodies when they come into contact under pressure. In the case of ball bearings, Hertzian contact theory explains how the balls or rollers deform slightly when pressed against the races (tracks) due to the load they support. This theory predicts the contact area and the distribution of stresses within the bearing, ensuring that the contact between the rolling elements and the races is optimized for minimal friction and wear while maintaining structural integrity.

Thus, by leveraging both Newton’s laws of motion and Hertzian contact theory, ball bearings can efficiently reduce friction and support loads in machinery. This ensures that the contact between the balls and races is optimized for minimal friction while maintaining structural integrity. Furthermore, ball bearings adhere to the first and second laws of thermodynamics, conserving energy and dissipating heat generated by friction through mechanisms such as lubrication. Overall, the functionality of ball bearings aligns with established scientific principles, making them essential in various mechanical applications.

While the initial choice of the omni-sphere for the puzzle project was scrapped due to cost effectiveness concerns the, the design was promising due to its strength, enriching design, and use of HDPE material. Furthurmore, the omni-sphere design was well-substantiated with STEM principles and practices, including the use of spherical geometry, material science, and mechanical engineering concepts such as ball bearings and friction reduction. Expert reviews confirmed the application of these principles, ensuring the validity and sustainability of the proposed solution for the puzzle project.

Our final design was nicknamed the spin drum. It consisted of a water drum with a removable lid. Attached to the inside of the lid by a lazy Susan is a PVC sheet with a hole cut out that lines up with a hole cut out in the lid. The design works by having the bear rotate the lazy Susan with a pulley device, such as a handle, rope, or the bear's claws, until the inner and outer holes align, allowing the bear to eat the food as it falls out of the hole.

The water drum we used was a standard drum that had been previously used by the zoo to entertain the bears. Given its prior use and proven durability, we determined it would be strong and safe enough for the bears, effectively eliminating the need for a material like HDPE, as we no longer needed to make the casing from scratch. HDPE was initially considered due to its high strength-to-density ratio, impact resistance, and non-toxic properties, making it ideal for creating safe and durable enclosures for animal interactions. However, repurposing the existing water drum provided a cost-effective alternative without compromising safety or functionality.

The use of a lazy Susan in the design leverages principles of mechanical engineering, particularly rotational mechanics. A lazy Susan operates on ball bearings, which facilitate smooth rotational motion by reducing friction between the moving parts. Ball bearings work based on the principles of rolling friction, which is significantly lower than sliding friction. By distributing the load evenly across the balls and races, the lazy Susan ensures that the rotation is smooth and requires minimal effort from the bear. This mechanism aligns with Newton's laws of motion, where the force applied by the bear is efficiently translated into rotational motion.

Furthermore, the choice of PVC for the internal sheet is grounded in material science principles. PVC, or polyvinyl chloride, is a widely used thermoplastic known for its durability, chemical resistance, and ease of fabrication. These properties make PVC an excellent choice for the internal mechanism, as it can withstand the mechanical stresses exerted by the bear's interactions and resist wear and tear over time. The ability to mold PVC into precise shapes also ensures that the alignment of the holes is accurate, maintaining the integrity of the puzzle mechanism.

The decision to use a drum instead of a sphere allowed us to simplify the design and reduce the need for an inner layer to prevent accidental treat spillage. The cylindrical geometry of the drum, combined with the strategically placed PVC layer, effectively contains the treats until the puzzle is solved. This design choice reflects principles of geometric optimization, where the simpler shape of the drum minimizes the complexity of construction while maximizing the functional efficiency of the device.

By incorporating these STEM principles into our design, the final product is not only more economical but also more robust and reliable. The integration of mechanical engineering concepts, material science, and geometric optimization ensures that the spin drum is a sustainable and valid solution. It retains the core interactive and enriching qualities intended for the bears while being practical for implementation. Additionally, the basic design of the spin drum allows for easy reuse and repair if it breaks, further emphasizing its sustainability and practicality for long-term use.

Overall, the spin drum project demonstrates a well-substantiated application of STEM principles and practices, addressing design requirements and functional claims effectively. The expert reviews confirmed the validity of our approach, ensuring that the spin drum provides an engaging and enriching activity for the bears while being efficient and feasible for the zoo to manage and maintain over time.